

App Features

CTP Job Cards Platform

Complete Feature Reference

A real-time digital maintenance management system. Explore every feature — from fault reporting to AI-assisted operations.

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App Overview

What CTP Job Cards does and how it fits into daily operations

CTP Job Cards is a **Flutter + Firebase** mobile application for field technician job card tracking. It targets Android (primary) with iOS and web as secondary platforms. The app operates offline-first — all data syncs automatically when connectivity is restored.

Core Value Pillars

- **Real-Time Response** — Fault reports reach the right technician in seconds. No dispatcher, no radio, no delay.
- **Full Accountability** — Every action is timestamped and logged. Who created it, who responded, when, what was done.

- **Intelligence Platform** — Structured data today powers predictive maintenance and AI fault-finding tomorrow.

End-to-End Process

- **Step 1 — Operator creates a job card.** Department, area, machine, component, fault description, priority, and type (Mechanical / Electrical / Mech-Elec) are captured and submitted.
- **Step 2 — System identifies on-site technicians.** Real-time geofencing determines which technicians of the correct trade are currently on site. Notifications are dispatched instantly — no human dispatcher required.
- **Step 3 — Technician accepts or responds.** From the notification: Assign Self (takes ownership, stops escalation), I'm Busy (acknowledges, system keeps looking), or Dismiss (logged — escalation continues).
- **If no response → Auto-Escalation.** Four configurable escalation stages fire. Stage 1 (5 min): Foremen and on-site managers. Stage 2 (10 min): Department managers. Stages 3 and 4 reserved for off-site management — disabled by default. Each stage is independently enable / disable-able by Admin.
- **Step 4 — Job tracked through to closure.** Status moves: Open → In-Progress → Monitor → Closed. Technician records what was done, parts used, root cause, and recommendations. Operator is notified on closure.

What's Live Today

Live:

- Job card creation & tracking
- Push notifications (P1–P5)
- Full-screen P5 alarms
- 4-stage auto-escalation
- Real-time geofencing
- Offline sync
- Permanent audit trail
- Manager dashboard
- Role-based access
- Copper inventory tracking
- Light & dark theme
- Notification action buttons

Planned:

- Predictive maintenance AI
 - AI chatbot & manuals
-

User Roles

Four roles with distinct access scopes and responsibilities

Every employee account in the system has a **role** field that controls what they can see and do. Roles are set by an Admin and are tied to each employee's clock number.

Operator

Discovers a fault and creates the job card. Responsible for capturing all details accurately at the point of report. Receives notifications when a technician accepts their job and when it is closed.

Key Screens: Create Job Card · Home · Job Card Detail

Technician

Receives job card notifications (when on site), attends to faults, self-assigns from the notification, and closes the job with a detailed completion note. Can be Mechanical, Electrical, or both.

Key Screens: Home · My Assigned Jobs · Job Card Detail · Create Job Card

Manager

Oversees all job cards in their department. Enforces data quality for operators and technicians. Receives escalation notifications. Has full department visibility on the Manager Dashboard.

Key Screens: Manager Dashboard · View Job Cards · Copper Dashboard · Notification History

Admin

Full system access. Manages employee accounts, configures geofence boundaries, adjusts escalation rules, and has access to all data across all departments.

Key Screens: Admin Screen · Geofence Editor · Employee Management

Authentication

All users sign in with their **Gmail address and password** (Firebase Auth). The system issues a custom token linked to each employee's clock number, which determines their role and department routing. Password resets are handled via email.

Operators and Technicians are distinct roles. A technician can also create job cards when they discover a fault, but operators are typically the frontline personnel who report issues. A technician receiving a job card they created would self-assign it automatically.

Job Cards

The core record unit — from fault report to closure

A job card is the formal digital record for every fault, breakdown, or maintenance task on site. It is the system's single source of truth — every subsequent action (notification, escalation, response, closure) is linked to a job card.

Job Card Fields

Field	Description	Why It Matters
Department	The plant department where the fault occurred	Routes notifications to the correct technicians
Area	Specific area within the department	Helps technicians locate the fault quickly
Machine	Named machine or asset ID	Enables asset-specific fault history and pattern analysis
Part / Component	The specific component affected	Precise identification for ordering parts and recording repairs
Description	What happened — symptoms, error codes, observations	Gives the technician context before arrival; feeds AI analysis
Priority	P1–P5 reflecting production impact	Determines notification type and escalation urgency

Field	Description	Why It Matters
Type	Mechanical / Electrical / Mech-Elec	Routes to the correct trade of technician
Closure Note	What was done, parts used, root cause, follow-up	Creates a permanent repair record; feeds future AI insight

Priority Levels

Priority	Production Impact	Expected Response	Notification Type
P1	No production effect — routine or planned	Attend when available	Standard banner
P2	Minor — production continuing	Attend soon	Standard banner
P3	Moderate — degraded operation	Within the shift	Standard banner
P4	Significant — output reduced	As soon as possible	Persistent banner
P5	Production standing — stopped	Immediate	Full-screen alarm + DND override

Priority must be honest. P5 means production has stopped. Misuse erodes the system's ability to distinguish genuine emergencies. Managers monitor priority accuracy — deliberate misuse must be corrected.

Job Card Status Flow

Open → In-Progress → Monitor → Closed

- **Open** — Created, no technician
- **In-Progress** — Technician self-assigned
- **Monitor** — Fault resolved, watching
- **Closed** — Complete & confirmed

Status	Meaning	Trigger
Open	Job is live, awaiting technician acceptance. Escalation is actively firing.	Automatic on creation
In-Progress	A technician has self-assigned and is actively working. Escalation stops immediately.	Technician taps "Assign Self"
Monitor	Fault resolved but machine is under observation for recurrence.	Technician updates status manually
Closed	Job fully complete. Closure note required. Operator notified. Record stored permanently.	Technician closes with note

My Assigned Jobs Screen

Technicians see all jobs they have accepted, sorted by priority. Tapping any job opens the full detail view with all fields, the fault description, and controls to update the status or add a closure note.

The priority badge colour-codes urgency at a glance — red for P5, orange for P4, amber for P3, green for P2, blue for P1.

- Pull-to-refresh updates the list in real time
- Jobs automatically reorder as priorities are updated
- Closed jobs are archived, not deleted

Notifications

Three distinct notification types scaled to fault priority

The notification system uses **Firestore Cloud Messaging (FCM)** with a native Kotlin implementation on Android. Each priority level triggers a different notification type — from a standard banner for routine jobs to a full-screen alarm that bypasses the lock screen for P5 critical faults.

Priority 1 · 2 · 3 — Normal Banner

Example: *New Job Card · P2 · Pump House — Cooling pump P-02 not priming — Line 3*

Action buttons: **Assign Self** · **I'm Busy** · **Dismiss**

Standard Android notification banner — appears at the top of the screen and can be swiped away. Uses the default notification tone. Tapping opens the job card directly.

Priority 4 — Persistent Banner

Example: *P4 Alert · Significant Impact — Line 2 press machine — reduced output · Workshop*

Action buttons: **Assign Self** · **I'm Busy** · **Dismiss**

Persistent banner — cannot be swiped away. Stays in the notification panel until an action button is tapped. Escalation alert tone with strong vibration. Prompt response required.

Priority 5 — Full-Screen Critical Alarm

Example: *PRODUCTION STOPPED — Conveyor 3 Drive Motor — Line 2 · P5 · Immediate Response Required*

Full-screen alarm — takes over the entire phone display, even from the lock screen. Implemented via native Android `FullScreenJobAlertActivity.kt` + `AlarmReceiver` . Powered by scheduled exact alarms so it fires even in deep sleep mode.

- Loud, repeating alarm tone
- Bypasses Do Not Disturb
- Appears over lock screen and other apps
- Cannot be passively ignored — requires a tap to dismiss
- Every response (or non-response) is logged with a timestamp

Notification Action Buttons

Every notification includes three action buttons that can be tapped without opening the app:

Action	Effect on Job Card	Effect on Escalation	Who Is Notified
Assign Self	Status → In-Progress. Technician assigned.	Stops immediately	Operator receives "Technician on the way"
I'm Busy	No change to status	Stops for this technician — system continues looking for another	Operator receives "Technician unavailable"

Action	Effect on Job Card	Effect on Escalation	Who Is Notified
Dismiss	No change	Escalation continues — next stage fires on schedule	Logged in notification history

On-Site Entry & Departure Notifications

The geofencing system also generates automatic notifications when technicians cross the site boundary — no user action required:

- **"Arrived On-Site"** — your on-site status updates and job card notifications begin routing to you
- **"Left Site Area"** — your status updates and notifications pause until you return

Automatic Escalation

Four configurable stages — no fault can be silently ignored

If a job card is not accepted by a technician, the escalation system automatically notifies progressively more senior recipients at timed intervals. Escalation is driven by **Cloud Functions** deployed to Google Cloud (`africa-south1`), running independently of any device or person.

Escalation stops **immediately** the moment any technician taps "Assign Self" or "I'm Busy" on any notification. If a job is escalated but then accepted, all further escalation stages are cancelled.

Default Escalation Timeline

- **T = 0 minutes — Job created — on-site technicians notified.** All on-site technicians of the correct trade type receive a push notification immediately.
- **Stage 1 · T = 5 minutes (default) — Foremen and on-site managers alerted — operator can also be opted in.** If no technician has responded within 5 minutes, foremen and on-site managers for the relevant department are notified with an escalation alert. The operator who raised the job can optionally be included — they receive a tailored "No response yet — please follow up directly" notification that tells them how many people have been notified so far.
- **Stage 2 · T = 10 minutes (default) — Department managers and workshop manager — urgent alert.** Escalated to management level. Managers receive a high-priority notification

indicating the job has been open without response for 10 minutes.

- **Stage 3 · T = 30 minutes (default — currently disabled) — Reserved for senior / off-site management.** Available but disabled by default. Once enabled by Admin, only jobs created from the enable moment onwards will trigger this stage — existing open jobs are protected from a notification flood.
- **Stage 4 · T = 60 minutes (default — currently disabled) — Final escalation tier.** The final configured escalation stage. All timers, recipient lists, and enable / disable toggles are configurable per stage by Admin under Settings → Escalation Config.

The times above are **defaults**, not fixed rules. An Admin can change the timer and recipient list for any stage at any time without a software update. Each stage can be enabled or disabled independently, and re-enabling a stage applies only to new jobs from that point forward (existing open jobs are not flooded).

Role-Based Routing by Trade

The Cloud Functions route notifications based on the job type:

- **Mechanical** jobs → Mechanics and mechanical foremen
- **Electrical** jobs → Electricians and electrical foremen
- **Mech-Elec** jobs → Both trades are notified simultaneously

Escalation recipients at Stage 1+ are department-specific — managers only see escalations from their own department, unless they are at senior management level.

Real-Time Geofencing

On-site status tracking — automatic, continuous, background

The geofencing system continuously monitors whether each technician is inside or outside the company boundary. Only **on-site technicians** receive job card notifications — off-site staff are excluded automatically. No one ever has to manually mark themselves as present or absent.

Legend:

- On site — receives notifications
- Off site — notifications paused

How It Works

- **Background Tracking** — The `BackgroundGeofenceService` runs continuously using Android WorkManager, even when the app is closed. It uses `geolocator` to monitor position relative to the configured site boundary polygon.
- **Battery Permission Required** — Android's battery optimiser can kill background services. Users must set the app to **Unrestricted** battery usage to ensure geofencing works overnight and during extended off-screen periods.
- **Entry & Exit Events** — The `GeofenceReceiver` fires on entry and exit. Each event updates the employee's Firestore document, sends an automatic on-site/off-site notification, and adjusts notification routing instantly.
- **Configurable Boundary** — Admins use the **Geofence Editor Screen** to define and adjust the site boundary polygon on a map. Changes take effect immediately for all users without a software update.

Location is not used for surveillance. The system records only whether an employee is inside or outside the configured site boundary — not their precise location or movement within the site.

Required Permission

Users must grant "**Allow All the Time**" location access (not "Only While Using the App"). Without background location access, geofencing cannot run when the screen is off, and notifications will fail to route correctly.

Copper Inventory Module

Real-time copper stock tracking and transaction logging

The copper inventory module provides a dedicated screen for tracking copper stock levels and recording transactions. It uses two Firestore collections — `copper_inventory` for current stock and `copper_transactions` for the transaction log.

Example stats:

Metric	Value
Current Stock	847 kg
This Week In	+120 kg
This Week Out	−65 kg

Example transactions:

- Stock received — Supplier A · +120 kg · Today 08:14
- Issued to Workshop B · −65 kg · Yesterday 14:30

Access

- The **Copper Dashboard** is accessible to Managers and Admins
- Inventory levels and transaction history are visible in real time
- All transactions are logged with timestamps, quantities, and responsible user

Offline Mode & Sync

Works without connectivity — syncs automatically when back online

CTP Job Cards is **offline-first**. If connectivity is lost, the app continues to function. All writes that cannot reach Firestore are queued locally in a Hive database and replayed automatically when the connection is restored.

Sync Flow

App Action (user creates/updates job) → Check Connectivity (connectivity_plus) → Queue Locally (Hive sync_queue) → Sync to Firestore (on reconnect)

How It Works

- **SyncService** initialises at app startup and listens to the `connectivity_plus` stream
- Failed writes (Firestore offline) are stored as `SyncQueueItem` objects in a Hive box named `sync_queue`

- When connectivity is restored, `SyncService` replays all queued items in order
- Firestore's built-in offline persistence also caches recent reads, so the UI remains usable even when offline

The sync queue uses **Hive** (a fast key-value store) for local persistence. `SyncQueueItem` is a `@HiveType` -annotated model — if you modify it, run `flutter pub run build_runner build` to regenerate the Hive adapters.

Local Storage Summary

Storage	Contents	Purpose
Hive — sync_queue	<code>SyncQueueItem</code> objects	Offline write queue — replayed on reconnect
SharedPreferences	<code>loggedInClockNo</code>	Persists the logged-in employee session
SharedPreferences	<code>permissionsCompleted</code>	Tracks whether onboarding permission flow has been completed
Firestore offline cache	Recent document snapshots	Read cache — UI works without network for cached data

App Permissions

Six permissions required for full functionality on Android

On first launch, the app guides users through a permission setup flow. All six permissions are required — each one enables a specific critical feature. The app will not function correctly if any are denied.

1. Location — "Allow All the Time"

Enables background geofencing so the system knows which technicians are on site. Without "All the Time" access, geofencing stops when the screen turns off — technicians miss notifications.

Tip: Must select "Allow All the Time" — not "Only While Using"

2. Notifications — "Allow"

All job card alerts reach the user through push notifications. Without this, no notifications are received and the system escalates as if no one is responding.

Tip: Always select "Allow"

3. Battery Optimisation — "Unrestricted"

Android's battery saver can kill the background geofencing service after a few minutes. Setting the app to Unrestricted prevents this — especially important overnight.

Tip: Settings → Apps → CTP Job Cards → Battery → Unrestricted

4. Display Over Other Apps

P5 full-screen alarms appear over the lock screen and above any other open app. Without this permission, P5 alerts fall back to a standard banner — easily missed.

Tip: Grant this permission for full P5 coverage

5. Do Not Disturb Access

Allows P5 critical alarms to bypass DND mode. If a technician has DND enabled overnight, P5 alarms must still sound for production-stopping faults.

Tip: Grant for full P5 alarm coverage

6. Schedule Exact Alarms

The P5 alarm uses the Android alarm system to guarantee exact-time firing, even in deep sleep mode. Without this, the alarm may arrive late or not fire at all.

Tip: Settings → Apps → Special App Access → Alarms & Reminders

Manager Dashboard

Live department visibility — job status, technician presence, escalation history

The Manager Dashboard gives department managers a real-time view of everything happening in their area. It is the primary tool for oversight, quality control, and escalation response.

Dashboard Panels

- **Live Job Overview** — All Open, In-Progress, Monitor, and recently Closed job cards for the manager's department, visible in real time as technicians update them.
- **Technician On-Site Status** — Which technicians are currently on site (and receiving notifications) versus off site, updated automatically by the geofencing system.
- **Escalation Log** — Jobs that have escalated — when, to whom, and what stage they reached. Recurring escalation patterns indicate coverage or response problems.
- **Notification History** — Full log of every notification sent — who received it, when, and what action they took. Provides accountability for non-response patterns.

Manager Responsibilities in the System

Responsibility	Why It Matters
Daily review of previous day's job cards	Catches missing records, slow responses, and emerging repeat failures
Enforce operator job card quality	Vague entries are worthless for analysis and future AI tools
Enforce technician closure note quality	A closure with no note is an incomplete maintenance record
Monitor escalation patterns	Repeated Stage 2+ escalations indicate a coverage or staffing problem
Identify machines with repeat faults	Same machine twice in 30 days warrants root cause investigation
Verify team on-site status accuracy	New phones or permission changes can break geofencing silently

Audit Trail

Permanent, append-only log of every action — searchable at any time

Every action taken on a job card — creation, status change, assignment, notification, escalation, closure — is written to the `job_card_audit` Firestore collection by **FirestoreService**. This is append-only: records are never modified or deleted.

Example Audit Entries

Time	Action	Actor
08:14:23	Job card created — P5 · Conveyor 3 Drive Motor · Line 2	J. Pretorius
08:14:25	Notifications dispatched — 4 on-site technicians (Mechanical)	System
08:15:10	Notification dismissed — escalation continues	K. Botha
08:16:25	Stage 1 escalation fired — foremen and on-site managers notified	System
08:17:02	Job assigned — status changed Open → In-Progress. Escalation cancelled.	D. van Wyk
09:03:47	Job closed — bearing replaced, motor restarted. Parts: SKF 6208-2RS. Root cause: age/fatigue.	D. van Wyk

What Is Captured

Data Point	Business Value
Machine, area, component	Precise asset identification for history tracking
Fault description	What happened and what was observed at the time
Priority at time of report	Production impact classification — searchable by priority level
Who created it, when	Operator accountability and fault timestamp
Who responded, when	Technician accountability and response time measurement (MTTR)
What was done, parts used	Repair record, parts consumption, cost basis
Root cause	Feeds predictive analysis and repeat-failure identification
Full notification log	Complete chain of who knew what and when — unalterable

Admin Tools

Full system control — accounts, geofences, escalation rules

Admins have access to all data across all departments plus system configuration tools not available to any other role.

- **Employee Management** — Create, edit, and deactivate employee accounts. Set roles, departments, and trade types. Link accounts to clock numbers.
- **Geofence Editor** — Draw and adjust the site boundary polygon on a live map. Changes take effect immediately for all users without a software update.
- **Escalation Rules** — Configure the timer and recipient list for each of the four escalation stages. Tune to match your site's staffing and response capacity.

Technology Stack

Component	Provider	Properties
Database	Google Firebase Firestore	Real-time, offline-capable, serverless, auto-scaling
Push Notifications	Firebase Cloud Messaging (FCM)	Enterprise-grade delivery, high-volume capable
Cloud Functions	Google Cloud Functions (africa-south1)	Serverless escalation logic — no server to maintain
Authentication	Firebase Auth	Email + password with custom tokens linked to clock numbers
Mobile App	Flutter (Dart)	Android primary; iOS and web capable; offline-first
Local Storage	Hive + SharedPreferences	Offline queue, session persistence

No on-premises infrastructure required. No server to maintain, no single point of hardware failure. All data is encrypted at rest and in transit. Backed by Google's 99.99% availability SLA with automatic scaling.

AI & Predictive Maintenance Roadmap

The current platform is a data foundation. The long-term value lies in what that data makes possible.

Phase 1 — Predictive Maintenance Analytics

Machine learning applied to job card history identifies failure patterns and predicts which assets are likely to fail — triggering proactive maintenance before breakdowns occur. Shifts the operation from reactive to predictive.

Phase 2 — AI Assistant — Manuals & Fault Finding

A conversational assistant integrated into the app gives technicians instant access to operator manuals, historical repair records, wiring diagrams, and parts specifications — in plain language, on their phone, on the floor.

Example AI Assistant Queries

- What are the most common causes of overload faults on Conveyor 3?
- What was done the last three times Machine 7 had a bearing failure?
- What is the correct torque spec for the FX400 gearbox drive shaft bearing?
- Which machines on Line 2 are statistically likely to fail in the next 30 days?

Business Case for AI

Benefit	Mechanism
Reduced unplanned downtime	Predictive alerts trigger maintenance before failure — not after
Faster fault resolution	AI-assisted diagnosis reduces time to identify and fix root cause
Knowledge retention	Institutional expertise captured in the system — not lost when experienced staff leave
Lower parts inventory cost	Predictive maintenance enables targeted stock — not blanket over-stocking
Reduced repeat failures	Root cause identified, addressed, and recorded — not just symptoms patched

Benefit	Mechanism
Technician upskilling	Access to manuals and history on-device reduces dependency on senior staff for routine questions

The quality of AI insight is entirely dependent on the quality of data entered today.

Every job card completed with accurate machine identification, clear fault descriptions, and detailed closure notes is a data point that improves predictive accuracy. Enforcing data standards at the management level is a direct investment in the long-term intelligence of the maintenance operation.